InsightGlobal

HEALTHCARE AI SOLUTION BLUEPRINTS

Five knowledge-graph blueprints across the healthcare value chain

IGH PRESIDENT BRIEFING · NOVEMBER 2026

Agenda

1. Who we are
2. Universal architecture
3. **CareGraph** · Care Engagement
4. **TherapyGraph** · Life Sciences
5. **RxGraph** · Pharmacy & Drug
6. **DeviceGraph** · Medical Devices & IoT
7. **ChainGraph** · Manufacturing & Supply Chain
8. How we engage

A core team built for any healthcare requirement

From strategy to production across AI, data, Salesforce and cloud, with healthcare compliance built in.

Agentic AI & GenAI engineering

Multi-agent workflows, RAG / GraphRAG, evaluation and guardrails, model-agnostic.

Knowledge graph & data engineering

Healthcare ontologies (RxNorm, ICD-10, SNOMED, LOINC, NPI, UDI), ETL, vector search.

Salesforce for healthcare

Health Cloud, Life Sciences Cloud, Marketing Cloud, Data Cloud, Agentforce, MuleSoft.

Multi-cloud delivery

Google Cloud, AWS and Azure. The same design runs wherever the client already is.

Healthcare compliance & governance

HIPAA, GxP / Part 11, FDA device and safety rules, consent, masking and audit trails.

Product & delivery

Ideation workshops, architecture, proof of concept to production, change management.

PROOF POINT We built a governed agentic delivery platform for a large US health system (Salesforce Health and Marketing Cloud, masked data context, compliance guardian, human approval gates).

Universal architecture

One architecture, five blueprints

Every blueprint uses the same four layers and trust layer, and runs on any cloud. Only the data, the central entity and the agents change.

DATA FOUNDATION

Sources we connect

- **Clinical systems** — EHR, claims (FHIR)
- **CRM & engagement** — Salesforce, contact center
- **Quality & operations** — QMS, MES, ERP
- **Devices & IoT** — Wearables, connected devices
- **Public reference data** — CMS, FDA, NLM, CDC
- **Documents & content** — Policies, SOPs, labels

DATA READINESS

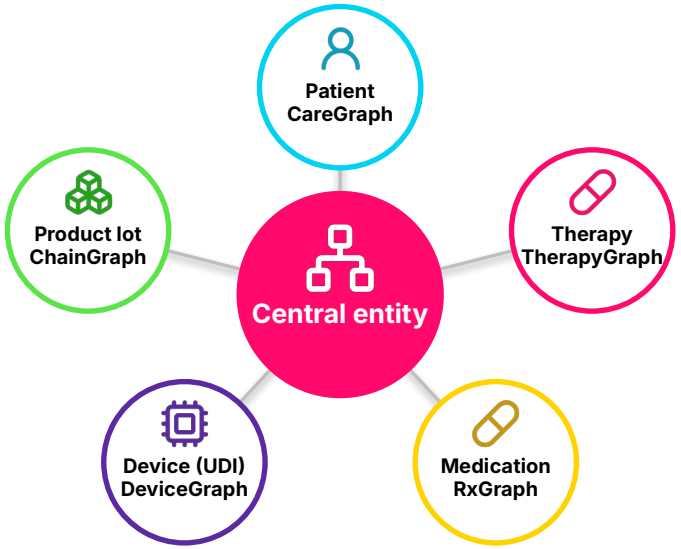
Matching, masking, codes

- **Match identities**
- **Map to standard codes**
- **Mask PHI & apply consent**
- **Make documents searchable**

INTELLIGENCE CORE

The domain model

FHIR · ICD-10 · RxNorm · UDI



AGENTIC WORKFLOWS

Agents + where users see it

Secure access
APIs · MCP tools

AI agent crew
model-agnostic

- Plan the task
- Retrieve facts
- Draft the output
- Check the rules
- Hand off to a person

Trust layer
PHI masking · guardrails in & out · human approval · audit trail

- **Salesforce**
- **Dashboards**
- **Ask in plain English**

Runs on

Google Cloud · AWS · Azure · Salesforce · Any LLM

Five blueprints across the healthcare value chain

Each blueprint is built around one central entity and solves the whole set of problems in its domain.

BLUEPRINT 01

CareGraph

CARE ENGAGEMENT

Center: **Patient**

- Cohort & gap intelligence
- Journey designer
- Access navigator
- Referral integrity
- Compliance guardian

BLUEPRINT 02

TherapyGraph

LIFE SCIENCES

Center: **Therapy**

- HCP 360 & next-best-action
- MLR pre-review
- Trial feasibility & sites
- Patient matching & recruitment
- Medical-affairs insight

BLUEPRINT 03

RxGraph

PHARMACY & DRUG

Center: **Medication**

- Medication reconciliation
- Interaction & PGx safety
- Benefits & prior auth
- Adherence & adverse events
- Shortage substitution

BLUEPRINT 04

DeviceGraph

MEDICAL DEVICES & IOT

Center: **Device (UDI)**

- Remote-monitoring triage
- Complaint & reportability
- Recall & UDI tracing
- Post-market trends
- Device cybersecurity

BLUEPRINT 05

ChainGraph

MANUFACTURING & SUPPLY CHAIN

Center: **Product lot**

- Supplier & material risk
- Deviation & CAPA copilot
- Batch-record review
- Inspection readiness
- Shortage early warning

THE PROBLEM

Care teams can't see, or act on, the whole patient journey

Care gaps, scheduling, referrals, costs and outreach sit in separate systems. Health systems run many campaigns but can't tell why a gap exists, what works for whom, or whether outreach is fair.

Who feels it: Population health, patient access, care coordination, marketing, quality

PROBLEMS IT SOLVES

Care gaps stay open
Screenings and chronic-care follow-ups are missed

One-size-fits-all outreach
Low conversion and high opt-outs

Hard to get an appointment
Long queues and abandoned scheduling

Referral leakage
Patients leave the network

Cost confusion
No clear estimates or financial help

Privacy slows everything
Each campaign waits on compliance review

EVIDENCE FROM THE FIELD

27%

better patient adherence reported by Salesforce Health Cloud customers
Source: Salesforce (vendor-reported)

OUR SOLUTION

CareGraph

Every patient journey, connected

One knowledge graph of the patient journey, with agents that find gaps, design outreach and book care, using masked data.

- **Cohort & gap intelligence** explains every gap by measure, community and access
- **Journey designer** personalized Marketing Cloud journeys
- **Access navigator** right provider, right slot, clear estimate
- **Referral integrity** keeps referrals in network
- **Compliance guardian** consent, HIPAA and TCPA checked before send

BUILT ON

Health Cloud · Marketing Cloud · Data Cloud · Agentforce · Tableau

CareGraph

Every patient journey, connected

On the universal architecture: Data Foundation › Data Readiness › Intelligence Core › Agentic Workflows

DATA FOUNDATION

Sources we connect

- **EHR / FHIR** — Encounters, problems, care gaps
- **CMS quality measures** — eCQM specifications
- **CDC PLACES + SDOH** — Community health by census tract
- **NPPES** — Provider identity (NPI)
- **Price transparency** — Hospital & payer rate files
- **Engagement data** — Marketing Cloud, contact center

DATA READINESS

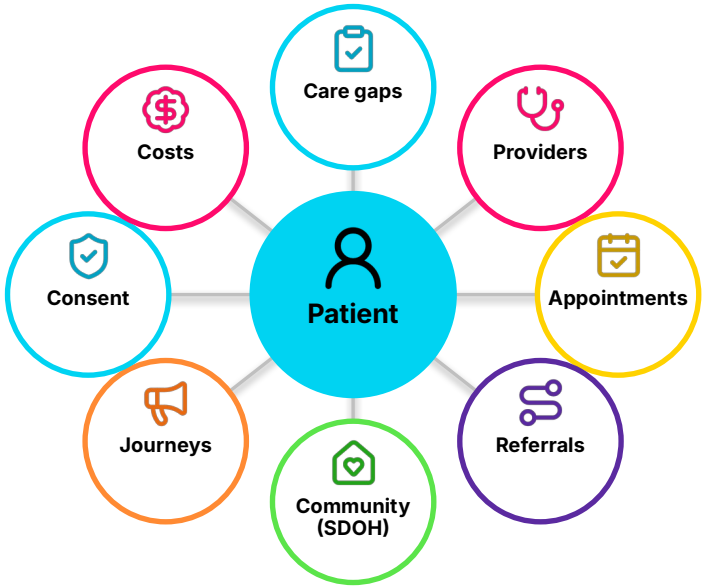
Matching, masking, codes

- Match patients & providers
- Map to standard codes
- Mask PHI & apply consent
- Make documents searchable

INTELLIGENCE CORE

The CareGraph model

FHIR · ICD-10 · LOINC · NPI



AGENTIC WORKFLOWS

Agents + where users see it

Secure access
APIs · MCP tools

AI agent crew
model-agnostic

- Cohort builder
- Gap explainer
- Journey designer
- Access navigator
- Compliance guardian

Trust layer
PHI masking · guardrails in & out · human approval · audit trail

- Health Cloud console
- Marketing Cloud journeys
- Ask CareGraph

Outcomes

- Care gaps closed
- Personalized outreach
- Visits booked
- Referrals kept
- Fair access
- Faster compliant campaigns

VALUE & BENEFITS

Close more care gaps, book more visits and keep patients in network, with every send compliant.

20–30%

higher care-gap closure rate per campaign

Pilot target (validated in pilot)

25–35%

fewer calls per booked appointment

Pilot target (validated in pilot)

29%

faster case resolution reported with Health Cloud

Source: Salesforce (vendor-reported)

100%

of outreach checked for consent and PHI before send

Design guarantee

Business

Quality-program revenue, more booked visits, less leakage

Clinical

Earlier screening and chronic-care follow-up

IT & data

One governed patient graph reused by every team

Compliance

Minimum-data access; every send auditable

THE PROBLEM

Commercial, medical and clinical teams work from different pictures

HCP data, trial data, publications and approved content live in separate systems. Brands struggle to personalize compliantly, MLR review takes weeks, and trials miss enrollment targets.

Who feels it: Brand and omnichannel teams, Medical Affairs, MLR, Clinical Operations

PROBLEMS IT SOLVES

Fragmented HCP view
CRM, claims and public data never meet

Slow MLR review
Every asset waits weeks for approval

Off-label risk
Claims drift from the approved label

Trial enrollment delays
Criteria are free text; screening is manual

Site selection by gut feel
Not grounded in patient data

Field insights get lost
MSL notes never reach strategy

EVIDENCE FROM THE FIELD

Oct 2025

Salesforce Life Sciences Cloud went GA; Pfizer and Fidia named early adopters, Novartis selected it globally

Source: Salesforce

OUR SOLUTION

TherapyGraph

One view of every therapy, from trial to market

One graph of each therapy's world (HCPs, evidence, trials and content) with agents for engagement, review and recruitment.

- **HCP 360 & next-best-action** channel, content and timing per HCP
- **MLR pre-review** flags off-label claims before human review
- **Trial feasibility & sites** masked patient counts by site and region
- **Patient matching & recruitment** every criterion explained, consent-aware outreach
- **Medical-affairs insight** summarizes field and literature signals

BUILT ON

Life Sciences Cloud · Marketing Cloud · Data Cloud · Agentforce

TherapyGraph

One view of every therapy, from trial to market

On the universal architecture: Data Foundation › Data Readiness › Intelligence Core › Agentic Workflows

DATA FOUNDATION

Sources we connect

- **NPPES** — HCP identity (NPI)
- **CMS Open Payments** — Industry payments to HCPs
- **Medicare Part D** — Prescribing by NPI and drug
- **ClinicalTrials.gov** — Trials, criteria, sites
- **DailyMed labels** — Approved indications & claims
- **PubMed** — Publications & authors

DATA READINESS

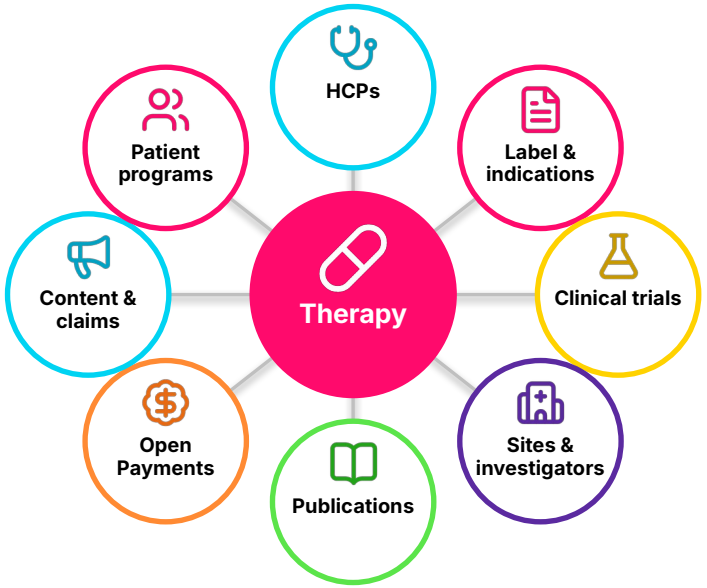
Matching, masking, codes

- **Match HCPs across sources (NPI)**
- **Map trials & labels to standard terms**
- **Apply consent & opt-outs**
- **Make content & publications searchable**

INTELLIGENCE CORE

The TherapyGraph model

NPI · NCT ID · SPL label · MeSH



AGENTIC WORKFLOWS

Agents + where users see it

Secure access
APIs · MCP tools

AI agent crew
model-agnostic

- Pre-call briefing
- Next-best-action
- MLR pre-review
- Trial feasibility
- Patient matcher

Trust layer
PHI masking · guardrails in & out · human approval · audit trail

- **Life Sciences Cloud console**
- **Marketing Cloud**
- **Ask TherapyGraph**

Outcomes

- Richer HCP engagement
- Faster MLR approval
- On-label content
- Faster trial enrollment
- Better site selection
- Field insight to strategy

VALUE & BENEFITS

More relevant HCP engagement, faster content approval and trials that enroll on time.

~40%

less trial screening time with NIH's TrialGPT approach

Source: NIH, Nature Communications 2024

30–50%

shorter MLR review cycle per asset

Pilot target (validated in pilot)

15–25%

higher HCP engagement rate

Pilot target (validated in pilot)

$14.67B

in HCP payments in CMS Open Payments (PY2025), linkable by NPI

Source: CMS via IntuitionLabs

Business

Faster launches, rep productivity, shorter trials

Clinical

Accurate, on-label information; more patients reach trials

IT & data

One therapy graph shared by commercial, medical and clinical

Compliance

Sunshine Act and PhRMA Code guardrails built in

THE PROBLEM

Medication decisions depend on data scattered across many systems

Medication lists are messy, interaction alerts are noisy, prior auth is manual, adherence slips and shortages hit without warning. General-purpose AI models also miss dangerous interactions.

Who feels it: Health-system and specialty pharmacy, PBMs, pharma patient-services hubs, drug safety

PROBLEMS IT SOLVES

Messy medication lists
Reconciled by hand at admission and discharge

Missed interactions
Noisy alerts hide real risks

Prior-auth backlogs
Packets assembled from charts by hand

Poor adherence
Refill gaps go unnoticed

Missed adverse events
Safety reports lost in support calls

Shortages
Substitutes chosen under pressure

EVIDENCE FROM THE FIELD

22.6K

free-text medication strings normalized to RxNorm by an LLM-assisted pipeline (RxMap)
Source: Johns Hopkins (peer-reviewed)

OUR SOLUTION

RxGraph

Every medication decision, grounded

One graph of every medication (ingredients, interactions, coverage, safety and supply) with agents that reconcile, check, authorize and substitute.

- **Medication reconciliation** free text to standard codes, discrepancies flagged
- **Interaction & PGx safety** graph lookup first; AI only explains
- **Benefits & prior auth** evidence mapped to payer criteria (CMS-0057-F ready)
- **Adherence & adverse events** refill nudges and safety-report capture
- **Shortage substitution** equivalent options checked against formulary

BUILT ON

Health Cloud | Service Cloud | Agentforce | EHR (FHIR)

RxGraph

Every medication decision, grounded

On the universal architecture: Data Foundation › Data Readiness › Intelligence Core › Agentic Workflows

DATA FOUNDATION
Sources we connect

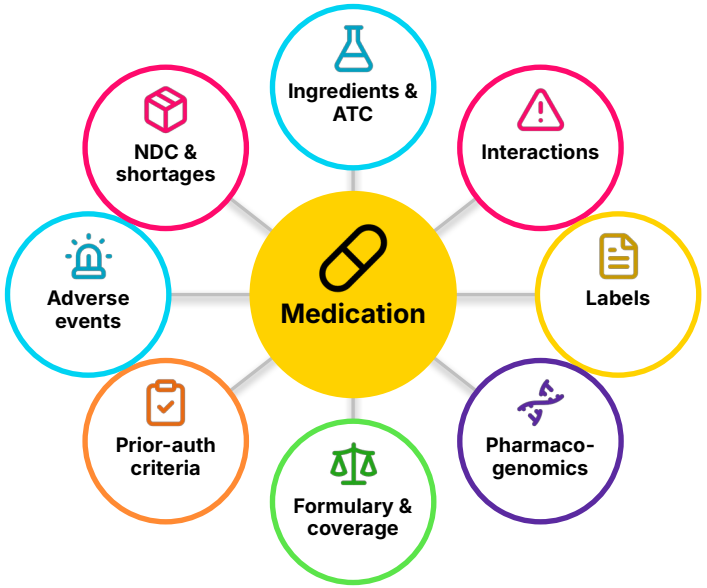
- **RxNorm** — Drug identity (RxCUI)
- **DailyMed / openFDA** — Labels, indications, warnings
- **openFDA FAERS** — Adverse event reports
- **CPIC / PharmGKB** — Gene–drug guidance
- **CMS Part D & LCD/NCD** — Formulary & coverage rules
- **FDA shortages & NDC** — Supply status

DATA READINESS
Matching, masking, codes

- **Normalize drug names to RxNorm**
- **Link labels, interactions & coverage**
- **Mask PHI & apply consent**
- **Make labels & policies searchable**

INTELLIGENCE CORE
The RxGraph model

Standards: RxNorm · NDC · ATC · FHIR



AGENTIC WORKFLOWS
Agents + where users see it

- **Secure access** — APIs · MCP tools
- **AI agent crew** — model-agnostic
  - Reconciler
  - Interaction checker
  - Prior-auth builder
  - Adherence coach
  - Substitute finder
- **Trust layer** — PHI masking · guardrails in & out · human approval · audit trail

→ **Pharmacist worklist**
→ **Health Cloud / EHR** (salesforce)
→ **Ask RxGraph**

Outcomes

- Safer medication lists
- Interactions caught
- Faster prior auth
- Better adherence
- Adverse events captured
- Shortages managed

VALUE & BENEFITS

Safer medication lists, faster access to therapy and fewer surprises from shortages.

4 of 4

AI models tested missed a known interaction, so answers must be grounded

Source: ERCSP 2024

40–60%

less pharmacist time per reconciliation

Pilot target (validated in pilot)

15–25%

higher prior-auth first-pass approval

Pilot target (validated in pilot)

Jan 2027

CMS prior-authorization API deadline (CMS-0057-F)

Source: CMS

Business

Fewer readmissions, faster time to therapy, less waste

Clinical

Fewer adverse drug events, continuity during shortages

IT & data

Standard drug codes and FHIR-ready prior-auth APIs

Compliance

Pharmacist approves every change; adverse events routed to safety

THE PROBLEM

Device data is everywhere, but nobody sees the whole picture

Remote-monitoring readings flood care teams, complaints pile up against tight FDA deadlines, recalls are hard to trace to patients, and connected devices carry known vulnerabilities.

Who feels it: MedTech quality & regulatory, virtual care / RPM programs, biomedical engineering

PROBLEMS IT SOLVES

Alert fatigue
Too many readings, too few real signals

Complaint backlogs
Manual triage and inconsistent coding

Late FDA reports
Reportability decided under time pressure

Recall traceability
Which patients have the affected device?

Device vulnerabilities
Known CVEs on connected devices

RPM billing missed
Data-day and minute rules tracked by hand

EVIDENCE FROM THE FIELD

4,459 colonoscopy device reports from FDA MAUDE classified by a GPT-4 pipeline with minimal human annotation

Source: Peer-reviewed study, IOS Press SHTI

OUR SOLUTION

DeviceGraph

Every device, traced from patient to recall

One graph that follows every device (UDI) from patient and telemetry to complaints, recalls and vulnerabilities, with agents for triage, reporting and remediation.

- **Remote-monitoring triage** rules and models decide; AI explains
- **Complaint & reportability** intake, IMDRF coding, FDA / EU MDR advice
- **Recall & UDI tracing** device to patient in one query
- **Post-market trends** early signals from MAUDE and internal data
- **Device cybersecurity** CVE-to-asset matching and patch plans

BUILT ON

Health Cloud · Service Cloud · MuleSoft · Tableau

DeviceGraph

Every device, traced from patient to recall

On the universal architecture: Data Foundation › Data Readiness › Intelligence Core › Agentic Workflows

DATA FOUNDATION

Sources we connect

- **Device telemetry** — Wearables & home devices (FHIR)
- **FDA GUDID** — UDI device identity
- **FDA MAUDE** — Device adverse events
- **FDA recalls & 510(k)** — Recalls, clearances, predicates
- **CISA ICSMA & NVD** — Device vulnerabilities
- **Complaints & service** — Service Cloud cases

DATA READINESS

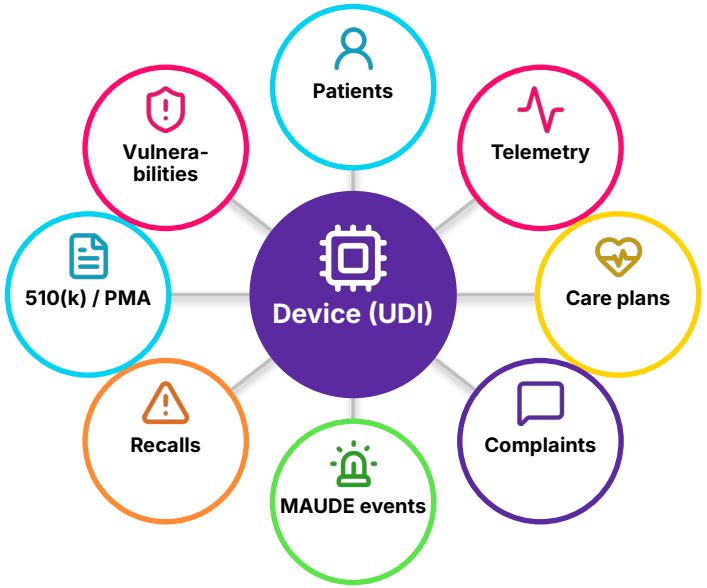
Matching, masking, codes

- **Pair devices to patients (UDI)**
- **Summarize telemetry daily**
- **Code events (IMDRF)**
- **Make narratives searchable**

INTELLIGENCE CORE

The DeviceGraph model

Standards: UDI · IMDRF · FHIR · CVE



AGENTIC WORKFLOWS

Agents + where users see it

- **Secure access** — APIs · MCP tools
- **AI agent crew** — model-agnostic
  - RPM triage
  - Complaint intake
  - Reportability advisor
  - Trend detector
  - Patch planner
- **Trust layer** — PHI masking · guardrails in & out · human approval · audit trail
- **Care team console**
- **Quality & regulatory**
- **Ask DeviceGraph**

Outcomes

- Fewer, better alerts
- Faster complaint handling
- On-time FDA reports
- Recalls traced
- Devices secured
- RPM revenue captured

VALUE & BENEFITS

Fewer, better alerts, faster complaint handling and devices you can trace and secure.

2M+

medical device reports FDA receives each year (MAUDE)

Source: FDA

30–50%

fewer alerts per nurse per day

Pilot target (validated in pilot)

30–40%

shorter complaint cycle time

Pilot target (validated in pilot)

40–60%

faster fix of critical device vulnerabilities

Pilot target (validated in pilot)

Business

Billable virtual care, lower cost of quality, fewer surprise recalls

Clinical

Earlier intervention and safety issues found sooner

IT & data

Standards-based device integration (FHIR, UDI)

Compliance

21 CFR 803, EU MDR vigilance; people make every reportability call

THE PROBLEM

From supplier to patient, the chain is only visible in pieces

Supplier risks are hidden, deviation investigations take weeks, repeat findings lead to FDA 483s, and shortages are discovered too late to plan around.

Who feels it: Pharma & biotech manufacturing, CDMOs, quality, supply chain, distributors

PROBLEMS IT SOLVES

Hidden supplier risk
Single-source materials and facilities

Slow investigations
Deviations start from a blank page

Repeat deviations
Root causes never connected; CAPAs overdue

Batch review bottleneck
eBMR review delays release

Inspection findings
Repeat FDA 483 themes

Late shortage warning
Stock-outs discovered too late

EVIDENCE FROM THE FIELD

55%

faster electronic batch-record review in a pharma deployment, with QA keeping final approval

Source: CannyMinds (vendor-reported)

OUR SOLUTION

ChainGraph

From supplier to patient, end to end

One graph of the product chain (supplier, material, plant, batch, distribution) with agents for quality, risk and shortage planning.

- **Supplier & material risk** single-source and facility risk map
- **Deviation & CAPA copilot** similar events, root cause and CAPA drafts
- **Batch-record review** exceptions flagged before QA release
- **Inspection readiness** matches findings to FDA 483 themes
- **Shortage early warning** supply signals and substitution plans

BUILT ON

Quality system (e.g. Veeva Vault QMS) | MES / LIMS / ERP | Tableau | Any cloud

ChainGraph

From supplier to patient, end to end

On the universal architecture: Data Foundation › Data Readiness › Intelligence Core › Agentic Workflows

DATA FOUNDATION

Sources we connect

- **ERP & suppliers** — Suppliers, materials, orders
- **MES / eBMR & LIMS** — Batches, equipment, tests
- **Quality system (QMS)** — Deviations, CAPAs, changes
- **FDA 483s & letters** — Inspection findings
- **FDA shortages & recalls** — Market supply signals (NDC)
- **DSCSA / distribution** — Serialization & inventory

DATA READINESS

Matching, masking, codes

- **Link supplier → lot → batch**
- **Map products & facilities (NDC)**
- **Validated, versioned pipelines**
- **Make SOPs & investigations searchable**

INTELLIGENCE CORE

The ChainGraph model

NDC · GS1 / DSCSA · GMP · ISO 13485



AGENTIC WORKFLOWS

Agents + where users see it

Secure access
APIs · MCP tools

AI agent crew
model-agnostic

- Supplier risk
- Deviation triage
- Root-cause drafter
- CAPA checker
- Shortage early warning

Trust layer
PHI masking · guardrails in & out · human approval · audit trail

- **Quality workbench**
- **Supply dashboard**
- **Ask ChainGraph**

Outcomes

- Supplier risk visible
- Faster investigations
- Fewer repeat deviations
- Faster batch release
- Inspection ready
- Shortages foreseen

VALUE & BENEFITS

Close investigations faster, stop repeat deviations and see shortages coming, with QA always in control.

40%

faster deviation closure reported in industry GenAI case studies

Source: McKinsey (vendor-reported)

323

active US drug shortages at the Q1 2024 peak

Source: ASHP

20–30%

fewer repeat deviations and overdue CAPAs

Pilot target (validated in pilot)

25–40%

earlier warning before a stock-out

Pilot target (validated in pilot)

Business

Right-first-time batches, less lost product, less emergency buying

Clinical

Consistent quality and continuity of supply for patients

IT & data

Validated, versioned pipelines (GAMP 5 / CSA)

Compliance

QA approves every AI draft; Part 11 audit trail

How we engage: from idea to production

Ideation workshop
2 WEEKS

Pick the blueprint and first modules, map data and stakeholders, agree success metrics.

Proof of concept
6–8 WEEKS

Working blueprint on public and synthetic data, architecture signed off, sponsor demo.

Pilot
~3 MONTHS

Real users, governed client data, measured against the agreed KPIs.

Scale
ONGOING

Add modules and sites, then the next blueprint on the same architecture.

Next step: pick one blueprint and run a 2-week ideation workshop with the client.

Sales cheat sheet: which blueprint to bring

Bring	Listen for...	Who to talk to
CareGraph Care Engagement	Care gaps, low screening rates, no-shows, outreach that doesn't convert, network leakage, price questions	CMO / CNO, VP Population Health, Patient Access, Marketing
TherapyGraph Life Sciences	Launch planning, HCP engagement, slow MLR review, trial enrollment delays, site selection	Brand / commercial leads, Medical Affairs, Clinical Operations
RxGraph Pharmacy & Drug	Medication errors, noisy alerts, prior-auth backlogs, CMS-0057-F, adherence, adverse events, shortages	Chief Pharmacy Officer, VP Patient Services, Drug Safety
DeviceGraph Medical Devices & IoT	Remote monitoring programs, alert fatigue, complaint backlogs, FDA / EU MDR reporting, recalls, device security	VP Quality / Regulatory, Post-Market Surveillance, Virtual Care, Biomed / CISO
ChainGraph Manufacturing & Supply Chain	Deviation backlogs, FDA 483s / warning letters, overdue CAPAs, supplier risk, drug shortages	VP Quality / QA, Site Heads, Supply Chain, Hospital Pharmacy Directors



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